Chapter 4 • Technical data - CompactFlash

1. General information

CompactFlash cards are easy-to-exchange storage media. Due to their robustness against environmental influences (e.g. temperature, shock, vibration, etc.), CompactFlash cards are ideal for use as storage media in industrial environments.

2. Basics

In order to be suited for use in industrial automation, CompactFlash cards must be highly reliable. To make this possible, the following is very important:

- Flash technology used
- Efficient algorithm for maximizing the lifespan
- Good mechanisms for detecting and fixing errors in the flash memory

2.1 Flash technology

Currently, CompactFlash cards are available with MLC (Multi Level Cell) and SLC (Single Level Cell) flash blocks.
SLC flash memory has a lifespan that is 10 times longer than MLC, which is why only CompactFlash cards with SLC flash blocks are suited for industrial applications.

2.2 Wear leveling

Wear leveling is an algorithm that can be used to maximize the lifespan of a CompactFlash card. There are three different algorithms:

- No Wear Leveling
- Dynamic Wear Leveling
- Static Wear Leveling

The basic idea behind wear leveling is to distribute data over a broad area of blocks or cells on the data carrier so that the same areas don't have to be cleared and reprogrammed over and over again.

2.2.1 No wear leveling

The earliest CompactFlash cards didn't have an algorithm for maximizing the lifespan. The lifespan of a CompactFlash card was determined only by the guaranteed lifespan of the flash blocks.

2.2.2 Dynamic wear leveling

Dynamic wear leveling makes it possible to utilize unused flash blocks when writing to a file.
If the data carrier is 80% full with files, then only 20% can be used for wear leveling.
The lifespan of the CompactFlash card is therefore dependent on the amount of unused flash blocks.

2.2.3 Static wear leveling

Static wear leveling also monitors which data is rarely changed. From time to time, the controller then moves this data to blocks that have already been frequently programmed in order to prevent further wear on those cells.

2.3 ECC error correction

Bit errors can be caused by inactivity or when a certain cell is operated. Error Correction Coding (ECC) implemented via hardware or software can detect and correct many errors of this type.

2.4 Maximum reliability

CompactFlash cards used by B&R use SLC flash blocks and static wear leveling together with a powerful ECC algorithm to provide maximum reliability.

3. CompactFlash cards 5CFCRD.xxxx-04

3.1 General information

> ### Information:
>
> **The 5CFCRD.xxxx-04 CompactFlash cards are supported on B&R devices with WinCE Version ≥ 6.0 or higher.**

3.2 Order data

Model number	Description	Figure
5CFCRD.0512-04	512 MB B&R CompactFlash card	
5CFCRD.1024-04	1024 MB B&R CompactFlash card	
5CFCRD.2048-04	2048 MB B&R CompactFlash card	
5CFCRD.4096-04	4096 MB B&R CompactFlash card	
5CFCRD.8192-04	8192 MB B&R CompactFlash card	
5CFCRD.016G-04	16 GB B&R CompactFlash card [1)]	
		CompactFlash card

Table 39: Order data - CompactFlash cards

1) In preparation

3.3 Technical data

Caution!

A sudden loss of power can cause data to be lost! In very rare cases, the mass memory may also become damaged.

To prevent damage and loss of data, it is recommended to use a UPS device.

Information:

The following characteristics, features and limit values only apply to this accessory and can deviate from those specified for the entire device. For the entire device where this accessory is installed, refer to the data provided specifically for the entire device.

Features	5CFCRD.0512-04	5CFCRD.1024-04	5CFCRD.2048-04	5CFCRD.4096-04	5CFCRD.8192-04
MTBF (at 25°C)	> 3,000,000 hours				
Maintenance	None				
Data reliability	< 1 unrecoverable error in 10^{14} bit read accesses				
Data retention	10 years				
Lifetime monitoring	Yes				
Supported operating modes	PIO Mode 0-6, Multiword DMA Mode 0-4, Ultra DMA Mode 0-4				
Continuous reading	Typically 35 MB/s (240X)[1][2] Max. 37 MB/s (260X)[1][2]	Typically 35 MB/s (240X)[1][2] Max. 37 MB/s (260X)[1][2]	Typically 35 MB/s (240X)[1][2] Max. 37 MB/s (260X)[1][2]	Typically 33 MB/s (220X)[1][2] Max. 34 MB/s (226X)[1][2]	Typically 27 MB/s (180X)[1][2] Max. 28 MB/s (186X)[1][2]
Continuous writing	Typically 17 MB/s (110X) Max. 20 MB/s (133X)	Typically 17 MB/s (110X) Max. 20 MB/s (133X)	Typically 17 MB/s (110X) Max. 20 MB/s (133X)	Typically 16 MB/s (106X) Max. 18 MB/s (120X)	Typically 15 MB/s (100X) Max. 17 MB/s (110X)
Endurance					
Guaranteed data volume [3] Results for 5 years [3]	50 TB 27.40 GB/day	100 TB 54,79 GB/day	200 TB 109,59 GB/day	400 TB 219,18 GB/day	800 TB 438,36 GB/day
Clear/write cycles Guaranteed Typical[4]	 100,000 2,000,000				
SLC flash	Yes				
Wear leveling	Static				
Error Correction Coding (ECC)	Yes				

Table 40: Technical data - CompactFlash cards 5CFCRD.xxxx-04

Support	5CFCRD.0512-04	5CFCRD.1024-04	5CFCRD.2048-04	5CFCRD.4096-04	5CFCRD.8192-04
Hardware	PP300/400, PPC700, PPC300, APC620, APC810, APC820				
Windows XP Professional	-	-	-	Yes	Yes
Windows XP Embedded	Yes	Yes	Yes	Yes	Yes
Windows CE 6.0	Yes	Yes	Yes	Yes	Yes 5)
Windows CE 5.0	-	-	-	-	-
PVI Transfer tool	≥ V3.2.3.8 (part of PVI Development Setup ≥ V2.06.00.3011)				
B&R Embedded OS Installer	≥ V3.0				
Mechanical characteristics					
Dimensions Length Width Thickness	 36.4 ±0.15 mm 42.8 ±0.10 mm 3.3 ±0.10 mm				
Weight	10 g				
Environmental characteristics					
Ambient temperature Operation Storage Transport	 0 to +70 °C -65 to +150 °C -65 to +150 °C				
Relative humidity Operation / Storage / Transport	Max. 85 % at 85 °C				
Vibration Operation / Storage / Transport	20 G peak, 20- 2000 Hz, 4 in each direction (JEDEC JESD22, method B103) 5.35 G RMS, 15 min per level (IEC 68-2-6)				
Shock Operation / Storage / Transport	1.5k G peak, 0.5 ms 5 times (JEDEC JESD22, method B110) 30 G, 11 ms 1 time (IEC 68-2-27)				
Altitude	Max. 15000 feet (4572 meters)				

Table 40: Technical data - CompactFlash cards 5CFCRD.xxxx-04 (cont.)

1) Speed specs with 1X = 150 KB/s. All specs refer to Samsung flash chips, CompactFlash cards in UDMA Mode 4, cycle time 30 ns in True-IDE Mode with sequential read/write test.

2) The file is written/read sequentially in True IDE mode with the DOS program Thruput.exe.

3) Endurance of B&R CF cards (linear written block size with ≥ 128 KB)

4) Depending on the average file size.

5) Not supported by B&R Embedded OS installer.

3.3.1 Temperature humidity diagram - Operation and storage

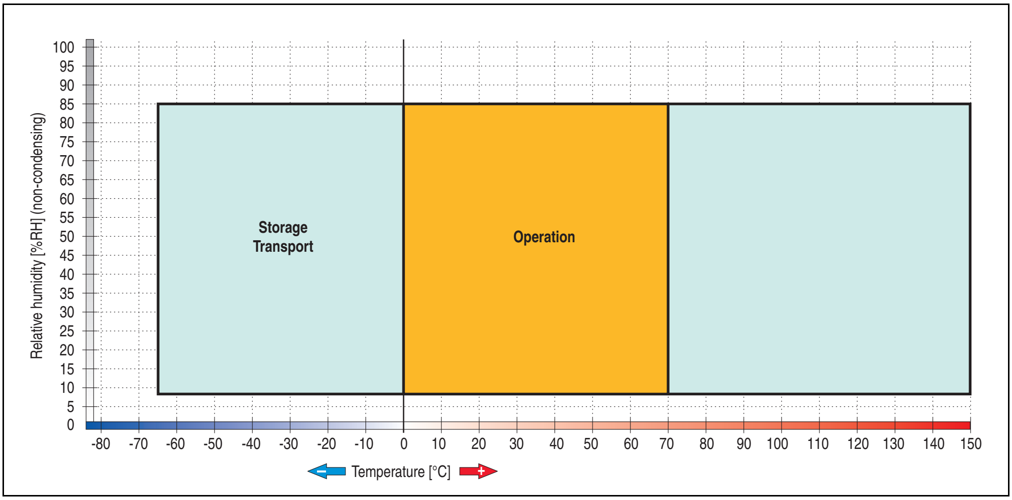

Figure 57: Temperature humidity diagram - CompactFlash cards 5CFCRD.xxxx-04

3.4 Dimensions

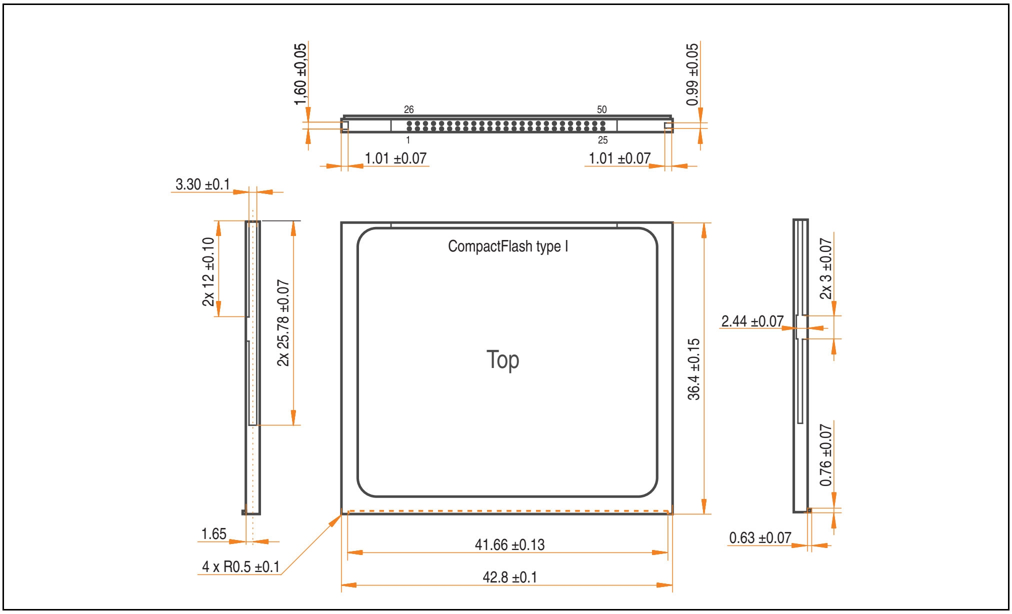

Figure 58: Dimensions - CompactFlash card Type I

3.5 Benchmark

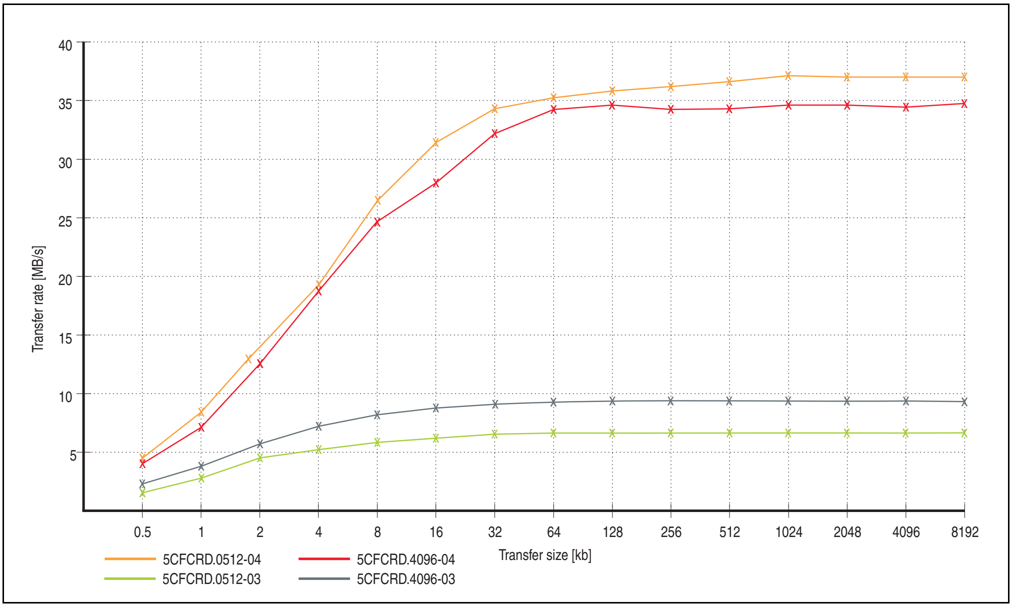

Figure 59: ATTO disk benchmark v2.34 comparison (reading)

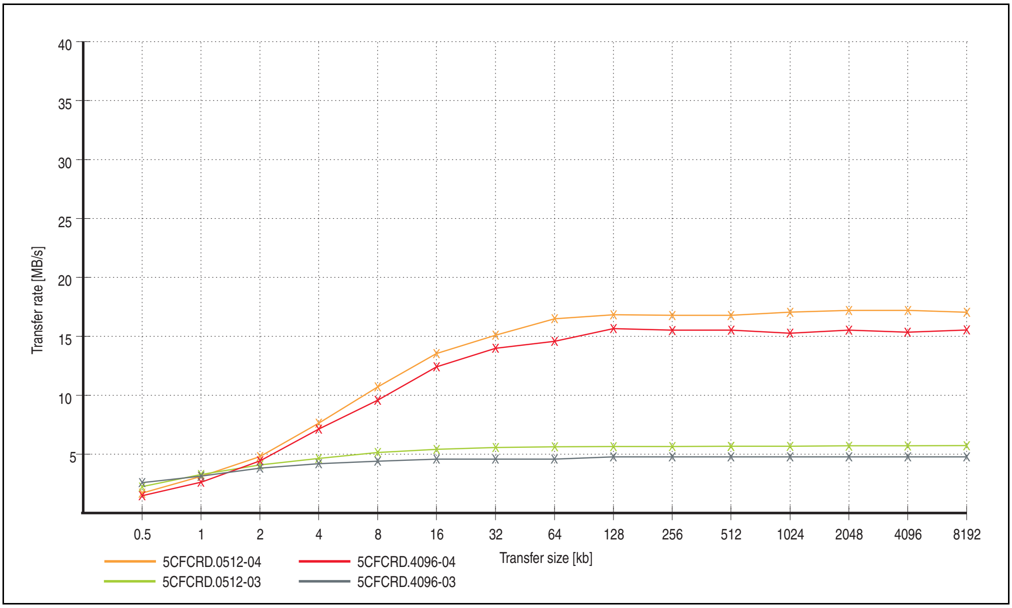

Figure 60: ATTO disk benchmark v2.34 comparison (writing)

4. CompactFlash cards - 5CFCRD.xxxx-03

4.1 General information

Information:

On Windows CE 5.0 devices, 5CFCRD.xxxx-03 CompactFlash cards up to 1GB are supported.

4.2 Order data

Model number	Description	Figure
5CFCRD.0064-03	CompactFlash 64 MB SSI	 CompactFlash card
5CFCRD.0128-03	CompactFlash 128 MB SSI	
5CFCRD.0256-03	CompactFlash 256 MB SSI	
5CFCRD.0512-03	CompactFlash 512 MB SSI	
5CFCRD.1024-03	CompactFlash 1024 MB SSI	
5CFCRD.2048-03	CompactFlash 2048 MB SSI	
5CFCRD.4096-03	CompactFlash 4096 MB SSI	
5CFCRD.8192-03	CompactFlash 8,192 MB SSI	

Table 41: Order data - CompactFlash cards

4.3 Technical data

Caution!

A sudden loss of power can cause data to be lost! In very rare cases, the mass memory may also become damaged.

To prevent damage and loss of data, B&R recommends that you use a UPS device.

Information:

The following characteristics, features and limit values only apply to this accessory and can deviate from the entire device. For the entire device where this accessory is installed, refer to the data provided specifically for the entire device.

Features	5CFCRD. 0064-03	5CFCRD. 0128-03	5CFCRD. 0256-03	5CFCRD. 0512-03	5CFCRD. 1024-03	5CFCRD. 2048-03	5CFCRD. 4096-03	5CFCRD. 8192-03
MTBF (at 25°C)	> 4,000,000 hours							
Maintenance	None							
Data reliability	< 1 unrecoverable error in 10^{14} bit read accesses							
Data retention	10 years							
Lifetime monitoring	Yes							
Supported operating modes	PIO Mode 0-4, Multiword DMA Mode 0-2							
Continuous reading	Typically 8 MB/s							
Continuous writing	Typically 6 MB/s							
Endurance								
Clear/write cycles Typical	> 2,000,000							
SLC flash	Yes							
Wear leveling	Static							
Error Correction Coding (ECC)	Yes							
Support								
Hardware	MP100/200, PP100/200, PP300/400, PPC700, PPC300, Provit 2000, Provit 5000, APC620, APC680, APC810, APC820							
Windows XP Professional	-	-	-	-	-	-	Yes	Yes
Windows XP Embedded	-	-	-	Yes	Yes	Yes	Yes	Yes
Windows CE 6.0	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes 1)
Windows CE 5.0	Yes	Yes	Yes	Yes	Yes	-	-	-

Table 42: Technical data - CompactFlash cards 5CFCRD.xxxx-03

Support	5CFCRD. 0064-03	5CFCRD. 0128-03	5CFCRD. 0256-03	5CFCRD. 0512-03	5CFCRD. 1024-03	5CFCRD. 2048-03	5CFCRD. 4096-03	5CFCRD. 8192-03
PVI Transfer tool	≥ V2.57 (part of PVI Development Setup ≥ V2.5.3.3005)							
B&R Embedded OS Installer	≥ V2.21							
Mechanical characteristics								
Dimensions Length Width Thickness	 36.4 ±0.15 mm 42.8 ±0.10 mm 3.3 ±0.10 mm							
Weight	11.4 g							
Environmental characteristics								
Ambient temperature Operation Storage Transport	 0 to +70 °C -50 to +100 °C -50 to +100 °C							
Relative humidity Operation / Storage / Transport	 8 to 95 %, non-condensing							
Vibration Operation Storage / Transport	 Max. 16.3 g (159 m/s² 0-peak) Max. 30 g (294 m/s² 0-peak)							
Shock Operation Storage / Transport	 Max. 1000 g (9810 m/s² 0-peak) Max. 3000 g (29430 m/s² 0-peak)							
Altitude	Maximum 80,000 feet (24,383 meters)							

Table 42: Technical data - CompactFlash cards 5CFCRD.xxxx-03 (cont.)

1) Not supported by B&R Embedded OS installer.

4.3.1 Temperature humidity diagram - Operation and storage

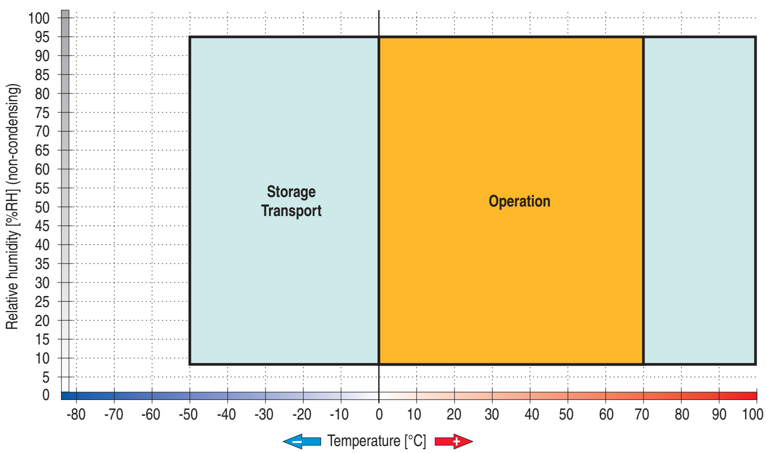

Figure 61: Temperature humidity diagram - CompactFlash cards 5CFCRD.xxxx-03

4.4 Dimensions

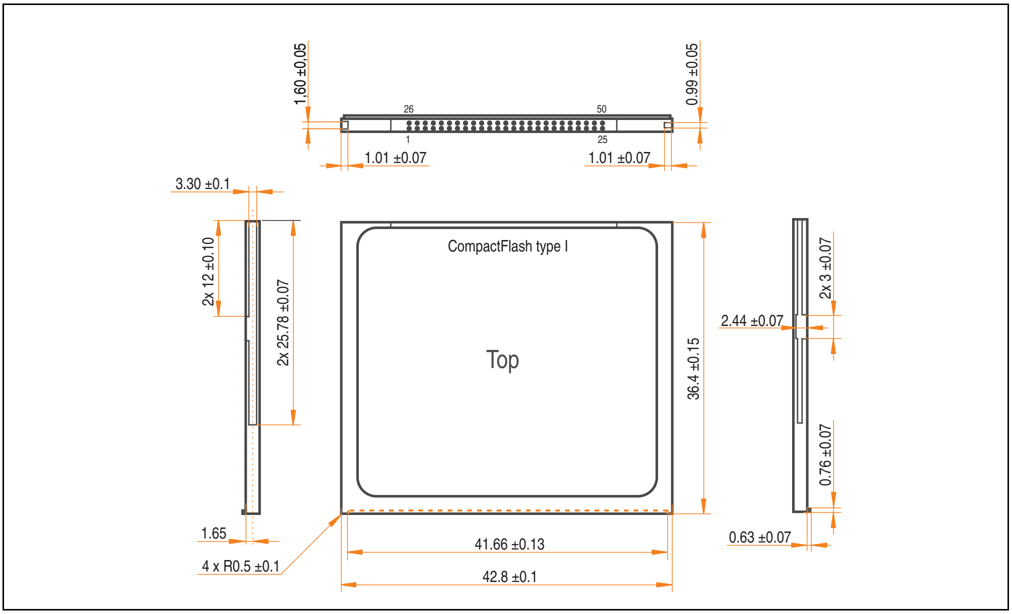

Figure 62: Dimensions - CompactFlash card Type I